

The relationship between advertising spending and sales

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1. Business Problem and Key Business Questions

Marketing budgets are finite, and firms must decide how to split spending across advertising channels to maximise sales. Allocating money to a channel that does not actually move sales wastes budget that could be redeployed to a channel that does. The business problem addressed in this report is: how is spending on TV, radio and newspaper advertising related to product sales across markets, and how should a fixed advertising budget be allocated across these three channels?

This translates into three key business questions:

1. Is spending on each individual channel (TV, radio, newspaper) significantly associated with sales, and how strong is each association?
2. When all three channels are considered simultaneously, which ones remain significant predictors of sales — i.e. which channels have an effect that is not simply explained by their correlation with another channel?
3. How much of the variation in sales can advertising spend explain overall, and how reliable is that explanation when tested on markets the model has not seen?

2. Dataset

The dataset used is the “Advertising” dataset from James et al. (2013), describing product sales in 200 different markets together with the advertising budget (in \$000s) allocated to TV, radio and newspaper in each market.

Source URL (raw CSV used in this project): <https://raw.githubusercontent.com/selva86/datasets/master/Advertising.csv> (original source: statlearning.com, James et al., 2013; documented at search.r-project.org’s glmtoolbox package page).

This dataset was not used in the module exercises, has exactly the simple, four-variable structure needed to focus on hypothesis testing and regression diagnostics without being obscured by data-cleaning complexity, and is well suited to the business problem because sales is a continuous outcome plausibly driven by three continuous advertising-spend predictors.

3. Hypotheses

Four hypotheses were pre-specified:

- H1: H0: the population correlation between TV spend and sales is zero. H1: it is non-zero (positive direction expected).
- H2: H0: the population correlation between radio spend and sales is zero. H1: it is non-zero (positive direction expected).
- H3: H0: the population correlation between newspaper spend and sales is zero. H1: it is non-zero (positive direction expected).
- H4 (multivariable): in a multiple linear regression of sales on TV, radio and newspaper spend simultaneously, H0: each individual regression coefficient is zero (that channel has no effect once the others are held constant). H1: at least one coefficient is non-zero. This tests whether each channel’s apparent bivariate relationship (H1-H3) survives once the other two channels are controlled for.

All tests were evaluated at $\alpha = 0.05$.

4. Data Preparation, Sampling and Cleaning

The raw file contains 200 rows and 5 columns (an unlabelled row-index column plus TV, radio, newspaper and sales). Inspection (`str()`, `colSums(is.na())`, `duplicated()`) found no missing values and no duplicate rows — the dataset is already clean, which is realistic for a curated teaching dataset but was verified rather than assumed. The redundant index column was dropped and variables were given consistent lower-case names. A plausibility check confirmed all spend and sales values are non-negative, as required for budget and unit-sales figures. The cleaned dataset retains all 200 markets, 4 variables.

```
df$X <- NULL
names(df) <- c("TV", "radio", "newspaper", "sales")
stopifnot(all(df >= 0))
```

Because the population is only 200 markets, the full dataset was used for every hypothesis test in Section 6, so no sub-sampling was required for statistical validity. To demonstrate sampling technique, and — more importantly — to check that the fitted regression model *generalises* rather than merely fitting the 200 markets on hand, an 80/20 simple random split (160 training markets, 40 test markets) was drawn. The model was fitted only on the training markets, and its predictive accuracy was evaluated on the held-out 40 test markets in Section 7.5.

5. Exploratory Data Analysis

Table 1 summarises the four variables.

Variable	Mean	SD	Min	Max
TV (\$000s)	147.04	85.85	0.7	296.4
Radio (\$000s)	23.26	14.85	0.0	49.6
Newspaper (\$000s)	30.55	21.78	0.3	114.0
Sales (000s units)	14.02	5.22	1.6	27.0

TV receives by far the largest and most variable budget allocation, while radio receives the smallest. The correlation matrix (Table 2) shows sales is most strongly correlated with TV ($r = 0.782$), moderately with radio ($r = 0.576$), and weakly with newspaper ($r = 0.228$). Notably, newspaper spend is itself correlated with radio spend ($r = 0.354$) — markets that spend more on radio also tend to spend more on newspaper — which foreshadows a confounding issue explored in Section 7.

	TV	radio	newspaper	sales
TV	1.000	0.055	0.057	0.782
radio	0.055	1.000	0.354	0.576
newspaper	0.057	0.354	1.000	0.228
sales	0.782	0.576	0.228	1.000

The TV relationship shows a clear, fairly tight positive linear trend; radio shows a positive but noisier trend; newspaper shows a weak, diffuse trend with the widest scatter around the line — consistent with the correlation coefficients above.

6. Test Selection, Assumption Checks and Justification

H1-H3 (bivariate associations). Both variables in each pair are continuous, so Pearson's correlation coefficient is the appropriate measure of linear association (Field, 2018). Pearson's test assumes approximately linear relationships (confirmed by the scatterplot review in Section 5) and is reasonably robust to non-normality at $n = 200$ by the Central Limit Theorem, so no separate normality pre-test was required for the correlation tests themselves.

H4 (multiple regression). Because TV, radio and newspaper spend are simultaneously present in each market and radio and newspaper are correlated with each other, only a multiple linear regression can isolate each channel's *independent* association with sales while holding the others constant (James et al., 2021). Ordinary least squares regression relies on four assumptions, each checked explicitly on the fitted model:

- Linearity: the residuals-vs-fitted plot shows a broadly random scatter around zero with a slight bow, indicating the linear specification is reasonable but not perfect (discussed as a limitation in Section 8).
- Normality of residuals: Shapiro-Wilk test on the regression residuals was significant ($W = 0.952$, $p < .001$), indicating a mild departure from normality (the Q-Q plot of residuals shows a few markets with unusually low sales for their spend, producing a modestly heavy left tail). With $n = 160$ in the training fit, the Central Limit Theorem still supports approximately valid t- and F-tests for the coefficients, but this is flagged as a limitation.
- Homoscedasticity: the Breusch-Pagan test was not significant ($BP = 2.65$, $df = 3$, $p = .448$), so the assumption of constant error variance is satisfied.
- Independence of errors: the Durbin-Watson test gave $D-W = 1.996$ ($p = .946$), showing no meaningful autocorrelation — expected, since each row is an independent market rather than a time series.
- Multicollinearity: variance inflation factors were all below 1.15 (TV = 1.00, radio = 1.14, newspaper = 1.15), far under the common threshold of 5 (James et al., 2021), so multicollinearity is not distorting the coefficient estimates despite the radio-newspaper correlation noted above.

7. Results

7.1 H1-H3 — Bivariate correlations with sales

```
cor.test(df$TV, df$sales); cor.test(df$radio, df$sales); cor.test(df$newspaper, df$sales)
```

All three channels were significantly, positively correlated with sales: TV $r = 0.782$, $t(198) = 17.67$, $p < .001$, 95% CI [0.722, 0.831]; radio $r = 0.576$, $t(198) = 9.92$, $p < .001$, 95% CI [0.475, 0.662]; newspaper $r = 0.228$, $t(198) = 3.30$, $p = .001$, 95% CI [0.092, 0.356]. H1, H2 and H3 are all supported: taken individually, every channel shows a statistically significant positive relationship with sales, though the strength differs sharply (TV largest, newspaper smallest).

7.2 H4 — Multiple linear regression (training set, $n = 160$)

```
model <- lm(sales ~ TV + radio + newspaper, data = train)
```

Predictor	Coefficient	Std. Error	t	p	95% CI
Intercept	3.012	0.331	9.09	< .001	[2.358, 3.667]

Predictor	Coefficient	Std. Error	t	p	95% CI
TV	0.0459	0.0015	30.67	< .001	[0.0429, 0.0488]
Radio	0.1888	0.0092	20.45	< .001	[0.1706, 0.2070]
Newspaper	-0.0055	0.0061	-0.90	.368	[-0.0176, 0.0066]

The overall model was highly significant ($F(3,156) = 472.7$, $p < .001$) and explained 90.1% of the variance in sales (adjusted R-squared = 0.899) — a very strong fit. TV and radio were both significant, positive predictors, but newspaper was not significant ($p = .368$) once TV and radio were held constant, even though newspaper showed a significant *bivariate* correlation in Section 7.1. This is explained by newspaper’s correlation with radio spend ($r = 0.354$, Section 5): newspaper’s apparent bivariate effect is largely borrowed from radio spend that tends to occur alongside it, not an independent effect of its own. Comparing standardised coefficients confirms the ranking of relative importance: TV (0.775) > radio (0.551) > newspaper (-0.024, non-significant).

H4 is therefore partially supported: TV and radio each have a non-zero independent effect on sales, but the null hypothesis of a zero coefficient for newspaper cannot be rejected.

7.3 Confirming newspaper’s redundancy: nested-model comparison

```
model_reduced <- lm(sales ~ TV + radio, data = train)
anova(model_reduced, model)
```

Removing newspaper had almost no effect on fit: R-squared fell only from 0.9009 to 0.9004, and an F-test comparing the two nested models was non-significant ($F(1,156) = 0.82$, $p = .368$ — identical to newspaper’s own t-test p-value, as expected for a single-predictor comparison). Newspaper spend adds no detectable explanatory power once TV and radio are in the model.

7.4 Out-of-sample validation (held-out 40 test markets)

Model	Test RMSE	Test R-squared
Full (TV + radio + newspaper)	1.953	0.876
Reduced (TV + radio only)	1.930	0.879

The two-predictor model generalises at least as well as the three-predictor model on markets the model never saw during fitting, reinforcing that newspaper spend can be dropped without any loss of predictive accuracy — indeed with a marginal gain, since a simpler model is less prone to overfitting noise.

8. Discussion, Recommendations, Limitations and Future Work

What the findings mean. Advertising spend as a whole explains sales extremely well (R-squared = 0.90), but the three channels are not equally responsible: TV has the largest independent effect, radio a clear secondary effect, and newspaper’s contribution disappears once TV and radio are accounted for. This directly answers the business questions: money spent on TV and radio moves sales; money spent on newspaper, in this data, does not have a detectable independent effect, and its simple correlation with sales appears to be an artefact of markets that spend on both radio and newspaper together.

Recommendations. (1) Continue prioritising TV, which has both the largest coefficient and the strongest bivariate correlation with sales. (2) Maintain or grow radio spend, which has a clear, statistically robust independent effect relative to its (currently smaller) budget share. (3) Reallocate newspaper budget toward TV or radio, or test this conclusion directly with a controlled experiment (e.g. a market-level A/B test reducing newspaper spend to zero in a random subset of markets) before eliminating it entirely, since observational data cannot fully rule out unmeasured confounders. (4) Because the residual diagnostics hinted at a slight non-linearity (Section 6), consider modelling a TV-by-radio interaction term in future analysis — a well-documented refinement of this exact dataset in the marketing-analytics literature — which may capture synergy between the two effective channels.

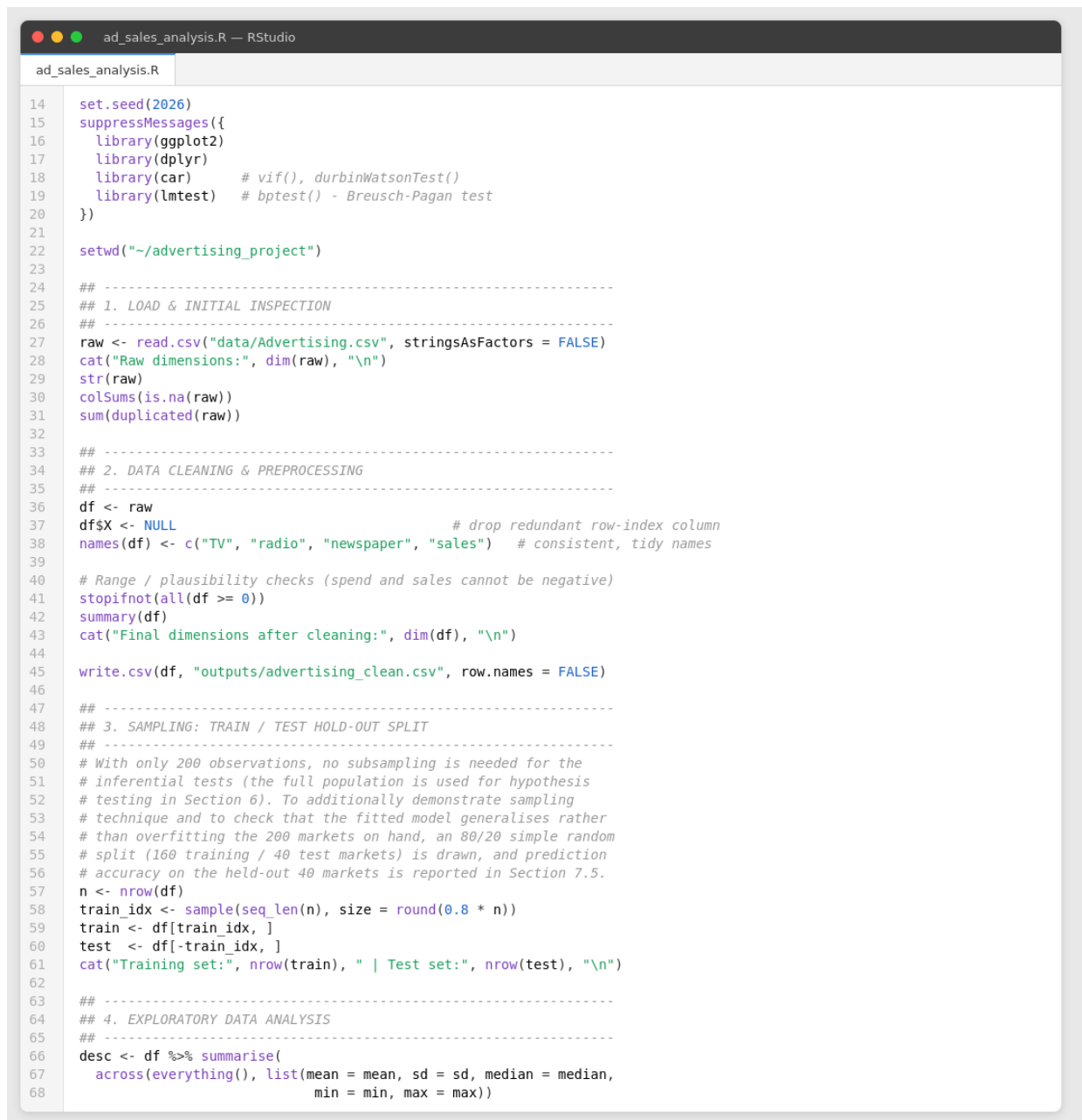
Limitations. The data are cross-sectional (200 markets at one point in time), so relationships describe *association*, not proven *causation*: unmeasured market characteristics (e.g. local competition, population size) could jointly influence both advertising allocation and sales. The residuals showed a mild, statistically significant departure from normality, suggesting the strictly linear, additive model is a good but imperfect approximation — a handful of markets sell less than the model predicts for their spend. The sample, while adequate for this analysis, is modest (200 markets), so the model’s precision for any single new market carries wider uncertainty than the aggregate R-squared suggests.

Future work. Fitting a model with a TV x radio interaction term, and comparing it by adjusted R-squared and out-of-sample RMSE against the additive model used here, would test directly whether the two effective channels reinforce each other. Extending the dataset with a time dimension (e.g. monthly spend and sales per market) would allow a distributed-lag or time-series model to capture delayed advertising effects, which a single cross-sectional snapshot cannot detect.

References (Harvard style)

- Field, A. (2018) *Discovering Statistics Using IBM SPSS Statistics*. 5th edn. London: Sage Publications.
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2013) *An Introduction to Statistical Learning with Applications in R*. New York: Springer, p.15.
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021) *An Introduction to Statistical Learning with Applications in R*. 2nd edn. New York: Springer.
- R Core Team documentation (2024) *advertising {glmtoolbox}*. Available at: <https://search.r-project.org/CRAN/refmans/glmtoolbox/html/advertising.html> (Accessed: 17 September 2026). Original data source: statlearning.com. Available at: <https://www.statlearning.com/s/Advertising.csv> (Accessed: 17 September 2026). Dataset mirror used in this project: selva86, Advertising.csv, GitHub. Available at: <https://raw.githubusercontent.com/selva86/datasets/master/Advertising.csv> (Accessed: 17 September 2026).

Appendix A: Full R Script (Screenshots)



```
ad_sales_analysis.R — RStudio
ad_sales_analysis.R

14 set.seed(2026)
15 suppressMessages({
16   library(ggplot2)
17   library(dplyr)
18   library(car)      # vif(), durbinWatsonTest()
19   library(lmtest)   # bptest() - Breusch-Pagan test
20 })
21
22 setwd("~/advertising_project")
23
24 ## -----
25 ## 1. LOAD & INITIAL INSPECTION
26 ## -----
27 raw <- read.csv("data/Advertising.csv", stringsAsFactors = FALSE)
28 cat("Raw dimensions:", dim(raw), "\n")
29 str(raw)
30 colSums(is.na(raw))
31 sum(duplicated(raw))
32
33 ## -----
34 ## 2. DATA CLEANING & PREPROCESSING
35 ## -----
36 df <- raw
37 df$X <- NULL          # drop redundant row-index column
38 names(df) <- c("TV", "radio", "newspaper", "sales") # consistent, tidy names
39
40 # Range / plausibility checks (spend and sales cannot be negative)
41 stopifnot(all(df >= 0))
42 summary(df)
43 cat("Final dimensions after cleaning:", dim(df), "\n")
44
45 write.csv(df, "outputs/advertising_clean.csv", row.names = FALSE)
46
47 ## -----
48 ## 3. SAMPLING: TRAIN / TEST HOLD-OUT SPLIT
49 ## -----
50 # With only 200 observations, no subsampling is needed for the
51 # inferential tests (the full population is used for hypothesis
52 # testing in Section 6). To additionally demonstrate sampling
53 # technique and to check that the fitted model generalises rather
54 # than overfitting the 200 markets on hand, an 80/20 simple random
55 # split (160 training / 40 test markets) is drawn, and prediction
56 # accuracy on the held-out 40 markets is reported in Section 7.5.
57 n <- nrow(df)
58 train_idx <- sample(seq_len(n), size = round(0.8 * n))
59 train <- df[train_idx, ]
60 test <- df[-train_idx, ]
61 cat("Training set:", nrow(train), " | Test set:", nrow(test), "\n")
62
63 ## -----
64 ## 4. EXPLORATORY DATA ANALYSIS
65 ## -----
66 desc <- df %>% summarise(
67   across(everything(), list(mean = mean, sd = sd, median = median,
68                               min = min, max = max))
68 )
```

Figure 1: R script, lines 14-68

```

ad_sales_analysis.R — RStudio
ad_sales_analysis.R
69 )
70 print(t(desc))
71 write.csv(t(desc), "outputs/descriptive_stats.csv")
72
73 cor_matrix <- cor(df)
74 print(round(cor_matrix, 3))
75 write.csv(round(cor_matrix, 3), "outputs/correlation_matrix.csv")
76
77 p1 <- ggplot(df, aes(TV, sales)) + geom_point(colour = "steelblue") +
78   geom_smooth(method = "lm", se = FALSE, colour = "darkred") + theme_minimal() +
79   labs(title = "Sales vs TV Advertising Spend", x = "TV spend ($000s)", y = "Sales (000s units)")
80 ggsave("outputs/fig1_sales_tv.png", p1, width = 5.5, height = 4, dpi = 150)
81
82 p2 <- ggplot(df, aes(radio, sales)) + geom_point(colour = "seagreen") +
83   geom_smooth(method = "lm", se = FALSE, colour = "darkred") + theme_minimal() +
84   labs(title = "Sales vs Radio Advertising Spend", x = "Radio spend ($000s)", y = "Sales (000s units)")
85 ggsave("outputs/fig2_sales_radio.png", p2, width = 5.5, height = 4, dpi = 150)
86
87 p3 <- ggplot(df, aes(newspaper, sales)) + geom_point(colour = "orange3") +
88   geom_smooth(method = "lm", se = FALSE, colour = "darkred") + theme_minimal() +
89   labs(title = "Sales vs Newspaper Advertising Spend", x = "Newspaper spend ($000s)", y = "Sales (000s units)")
90 ggsave("outputs/fig3_sales_newspaper.png", p3, width = 5.5, height = 4, dpi = 150)
91
92 p4 <- ggplot(df, aes(sales)) + geom_histogram(bins = 15, fill = "grey40") +
93   theme_minimal() + labs(title = "Distribution of Sales", x = "Sales (000s units)")
94 ggsave("outputs/fig4_sales_hist.png", p4, width = 5.5, height = 4, dpi = 150)
95
96 ## -----
97 ## 5. INFERENCE TESTS: SIMPLE CORRELATIONS (H1-H3)
98 ## -----
99 cor_tv <- cor.test(df$TV, df$sales)
100 cor_radio <- cor.test(df$radio, df$sales)
101 cor_news <- cor.test(df$newspaper, df$sales)
102 print(cor_tv); print(cor_radio); print(cor_news)
103
104 ## -----
105 ## 6. MULTIPLE LINEAR REGRESSION (H4) + ASSUMPTION CHECKS
106 ## -----
107 model <- lm(sales ~ TV + radio + newspaper, data = train)
108 summary(model)
109 confint(model)
110
111 # 6.1 Linearity: fitted-vs-residual plot should show no pattern
112 png("outputs/fig5_residuals_fitted.png", width = 700, height = 500, res = 150)
113 plot(model, which = 1)
114 dev.off()
115
116 # 6.2 Normality of residuals
117 shapiro.test(resid(model))
118 png("outputs/fig6_qq_residuals.png", width = 700, height = 500, res = 150)
119 plot(model, which = 2)
120 dev.off()
121
122 # 6.3 Homoscedasticity: Breusch-Pagan test
123 bptest(model)

```

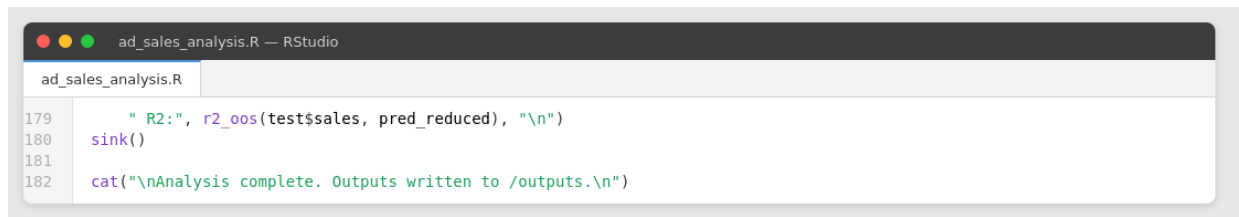
Figure 2: R script, lines 69-123

```

ad_sales_analysis.R — RStudio
ad_sales_analysis.R
124 # 6.4 Independence of errors: Durbin-Watson test
125 durbinWatsonTest(model)
126
127 # 6.5 Multicollinearity: Variance Inflation Factor
128 vif(model)
129
130 ## -----
131 ## 7. INTERPRETING THE MODEL
132 ## -----
133 # 7.1 Overall model significance already in summary() (F-statistic)
134 # 7.2 Standardised coefficients, to compare relative importance
135 train_std <- as.data.frame(scale(train))
136 model_std <- lm(sales ~ TV + radio + newspaper, data = train_std)
137 cat("Standardised coefficients:\n"); print(coef(model_std))
138
139 # 7.3 Simple-vs-multiple regression comparison for newspaper
140 model_news_only <- lm(sales ~ newspaper, data = train)
141 summary(model_news_only)
142
143 # 7.4 Model excluding the non-significant newspaper predictor
144 model_reduced <- lm(sales ~ TV + radio, data = train)
145 summary(model_reduced)
146 anova(model_reduced, model) # nested-model F-test: does newspaper add explanatory power?
147
148 # 7.5 Out-of-sample validation on the held-out 20% test set
149 pred_full <- predict(model, newdata = test)
150 pred_reduced <- predict(model_reduced, newdata = test)
151 rmse <- function(actual, pred) sqrt(mean((actual - pred)^2))
152 r2_oos <- function(actual, pred) 1 - sum((actual-pred)^2) / sum((actual-mean(actual))^2)
153 cat("Full model - Test RMSE:", round(rmse(test$sales, pred_full),3),
154     " | Test R2:", round(r2_oos(test$sales, pred_full),3), "\n")
155 cat("Reduced model- Test RMSE:", round(rmse(test$sales, pred_reduced),3),
156     " | Test R2:", round(r2_oos(test$sales, pred_reduced),3), "\n")
157
158 ## -----
159 ## 8. SAVE KEY NUMERIC RESULTS FOR THE REPORT
160 ## -----
161 sink("outputs/model_summary.txt")
162 cat("=== Pearson correlations with sales ===\n")
163 print(cor_tv); print(cor_radio); print(cor_news)
164 cat("\n=== Full multiple regression (training set) ===\n"); print(summary(model))
165 cat("\n=== 95% CIs for coefficients ===\n"); print(confint(model))
166 cat("\n=== Assumption checks ===\n")
167 cat("Shapiro-Wilk (residuals):\n"); print(shapiro.test(resid(model)))
168 cat("Breusch-Pagan:\n"); print(bptest(model))
169 cat("Durbin-Watson:\n"); print(durbinWatsonTest(model))
170 cat("VIF:\n"); print(vif(model))
171 cat("\n=== Reduced model (TV + radio only) ===\n"); print(summary(model_reduced))
172 cat("\n=== Nested model F-test (newspaper's added value) ===\n")
173 print(anova(model_reduced, model))
174 cat("\n=== Out-of-sample test-set performance ===\n")
175 cat("Full model RMSE:", rmse(test$sales, pred_full),
176     " R2:", r2_oos(test$sales, pred_full), "\n")
177 cat("Reduced model RMSE:", rmse(test$sales, pred_reduced),
178

```

Figure 3: R script, lines 124-178

A screenshot of the RStudio interface. The title bar at the top reads "ad_sales_analysis.R — RStudio". Below the title bar, the script editor shows the file "ad_sales_analysis.R". The script content is as follows:

```
179     " R2:", r2_oos(test$sales, pred_reduced), "\n")
180   sink()
181
182   cat("\nAnalysis complete. Outputs written to /outputs.\n")
```

Figure 4: R script, lines 179-182

Appendix B: GitHub Repository

https://github.com/Anashwar23/B105_Applied_Stat